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ENVIRONMENTAL IMPACT ASSESSMENT
OF
1200 MW COAL-BASED THERMAL POWER PLANT , VILE BHAGAD MIDC

COURSE NAME:

ENVIRONMENTAL AND SOCIAL IMPACT ASSESSMENT

COURSE CODE: HS-11

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1. Description of the Site

The proposed 1200 MW coal-fired thermal power station is situated in Vile Bhagad MIDC, Raigad district, Maharashtra, near to an ecologically sensitive area renowned for its biodiversity and distinctive natural sceneries.

1.1. GEOGRAPHICAL AND INFRASTRUCTURAL CONTEXT

1.1.1. ***Topography***: The property features gently sloping terrain.

1.1.2. ***Industrial Development***: Located within the Maharashtra Industrial Development Corporation (MIDC) zone, an officially defined industrial sector.

1.1.3. ***Connectivity***: Proximate to National Highway 66, facilitating effective road transportation for raw materials and final products. Strategically linked to railway networks for the transportation of resources, especially coal.

1.1.4. ***Hydrology***: The Kundalika River, an important hydrological entity, flows through the area, 3.5 km to the north.

1.1.5. ***Precipitation***: The area has significant monsoon precipitation, which substantially impacts local drainage systems.

<u>Pic 1. Satellite image of Vile Bhagad</u>

1.2. CLIMATIC AND METEOROLOGICAL VARIABLES

1.2.1. Wind Direction:

Mainly from the west. It is a critical factor for planning emission dispersion and air quality management strategies.

1.2.2. Seasonal Variations:

Intense monsoons create temporary water bodies and a fluctuating hydrological profile, affecting both natural and industrial ecosystems.

1.2.3. Air Quality:

- ***i.*** ***Total Suspended Particulate Matter (TSPM)***: 251.3 microgrammes/cubic metre (Maximum)
- ***ii.*** ***Respirable Particulate Matter (RPM)***: 66.8 µg/m³
- ***iii.*** ***SO2***: 21.6 microgrammes/cubic metre
- ***iv.*** ***NOx***: 24.6 microgrammes/cubic meter

1.2.4. Water Quality:

Compliance with IS 2296 & 10500: 2012 requirements for potable water quality.

- ***i.*** ***Surface water quality***: pH 7.3, Total Dissolved Solids: 163-175 mg/L
- ***ii.*** ***Groundwater Quality***: pH: 7-7.1, Total Dissolved Solids: 429-468 mg/L

1.2.5. **Soil Quality:** Moderate Fertility

- ***i.*** ***pH (7.1-7.3)***: Primarily Silt Clay
- ***ii.*** ***Nitrogen***: 121-143 kg/ha
- ***iii.*** ***Phosphorus***: 30-41.5 kg/ha
- ***iv.*** ***Potassium***: 205-244 kg/ha

1.3. ECOLOGICAL AND BIODIVERSITY CONTEXT

The region is located 4.5-5 km west of the Western Ghats, a globally recognised biodiversity hotspot, encompassing a blend of industrial and natural landscapes.

1.3.1. Habitat:

Grasslands at low to mid-elevations interspersed with semi-evergreen and deciduous woodlands. Seasonal fluctuations generate temporary inundations and pools that sustain plant and animal species.

1.3.2. **Diversity of Flora:**

i. A total of 98 species from 43 families were discovered, comprising 65 herbaceous species, 22 tree species, 6 shrub species, and 5 climbing species.

ii. ***Predominant families***: Fabaceae (15 species) and Malvaceae (8 species). *Erinocarpus nimmonii*, categorized as "Rare Endangered."

1.3.3. **Biodiversity of Fauna:**

i. ***Mammals***: 54 species distributed across 24 families (e.g., Muridae: 7 species, Vespertilionidae: 6 species).

ii. A total of 104 bird species were sighted, predominantly from the families Muscicapidae (11 species) and Accipitridae (7 species).

iii. ***Schedule IV***: 78 species.

iv. ***Schedule I***: Three species.

v. 66 fish species (7.5% vulnerable, 6% near threatened, 56% least concern, 3% data deficient, and 26% species status not available.

vi. Additional fauna comprises reptiles, amphibians, and arthropods (e.g., butterflies, dragonflies).

1.4. PROXIMITY TO SENSITIVE AREAS

1.4.1. **Western Ghats**: Located adjacent to this ecologically sensitive area.

1.4.2. **Tamhini Wildlife Sanctuary (WLS):** Situated 4.7 miles east of the location, with the closest Eco-Sensitive Zone (ESZ) settlement 4.5 km away.

1.4.3. **Villages identified as Ecologically Sensitive Areas (ESA):** Bhuvan, Vawe Diwali, Bondshet, and Kumbharte, are in proximity, highlighting the imperative for prudent development.

1.4.4. **Coastal Regulation Zone (CRZ) Limitations:** Situated in the area, requiring compliance with legal safeguards.

1.5. OVERVIEW OF SEISMIC AND GEOHAZARDS

1.5.1. The area is designated as Zone-IV (High Damage Risk Zone) on the Seismic Zonation of the Earthquake Hazard Map (BMTPC).

1.5.2. The site is not located in a landslide-prone area as per the Landslide Incidence Map (BMTPC).

1.5.3. Flood Hazard: The Flood Hazard Map (BMTPC) indicates that the area is not susceptible to flooding.

1.5.4. The site is designated as a Moderate Damage Risk Zone according to the Wind Hazard Map (BMTPC).

1.6. SEASONAL OBSERVATIONS

1.6.1. **Monsoon:**

Increased amphibian populations and seasonal vegetation. Ephemeral puddles promote biodiversity, including transient flora, odonates and amphibians.

1.6.2. **Winter:**

Reduced amphibian activity due to arid environments. Avian diversity attains its peak, exploiting seasonal habitats like the Kalvan and Dakhane plateaus.

2. Description of the Project

Feature	Description
Total Capacity	1200 MW
Configuration	2 x 600 MW
Type of Boilers	Pulverized Coal Fired
Fuel Source	Coal
Annual Coal Requirement	7.5 MTPA (Approx. 21, 865 TPD)
ESP Efficiency	99.9%
Stack Height	275 m
Water Requirement	3,400 m^3/hr
Power Evacuation	Via 400 kV switchyard to national grid

2.1. The plant will use high-efficiency pulverized coal-fired boilers that produce steam at 16.7 MPa at 538°C to drive condensing turbo-generator units.

2.2. Main coal source will be extracted coal from the nearby reserves at approximately 7.5 MTPA to help provide a Plant Load Factor (PLF) of around 93%.

2.3. The auxiliary facilities of this plant are coal and ash handling, a water treatment facility, cooling water systems, and ESPs (Electrostatic Precipitators) for air quality.

2.4. A 275-metre twin flue stack will be built to provide optimum dispersion of pollutants, and provisions for future FGD (Flue Gas Desulfurization) unit installation.

2.5. The facility will require about 3,400 cubic meters of water every hour, and this water will be obtained from the local sources while technologies of wastewater treatment ensure compliance with zero liquid discharge regulations.

2.6. Effective strategies for the collection and disposal of fly ash and bottom ash will be undertaken in compliance with environmental regulations.

3. Environmental Impacts

3.1. CONSTRUCTION AND PRECONSTRUCTION PHASE

3.1.1. Site clearing and levelling

i. ***Impact on air*:** the activities associated with this resulted in fugitive dust emission and noise pollution from the equipment used, this can lead to degradation in air quality, can cause respiratory issues in humans, and disturbances to the wildlife.

ii. ***Impact on water*:** Runoff from stripped vegetation and construction areas will contaminate the water bodies, reducing water availability and harming the aquatic ecosystem and biodiversity.

iii. ***Impact on soil*:** The fertile topsoil will be lost in the process, and alterations in drainage patterns can have a long-term impact on agricultural productivity. They can increase the risk of soil erosion. Removing vegetation and topographic changes can destroy the habitat, disrupt local ecosystems and accelerate soil erosion.

3.1.2. Transportation and storage of construction materials

i. ***Impact on air*:** The transportation of materials will generate fugitive dust, and there will be noise from vehicular movement. Further, it will degrade air quality and cause stress to nearby residents and the wildlife.

ii. ***Impact on land and water*:** the spillage of construction meterials to surroundings will contaminate soil, reduce its fertility, and can potentially impact the groundwater quality. When construction materials run off into streams or water storage areas, water sources become polluted, making them unfit for drinking or irrigation and posing risks to aquatic ecosystems.

3.1.3. Civil Construction Activities

i. ***Impact on air*:** Civil construction activities will emit significant noise, fugitive dust and will release air pollutants from machinery. This will reduce the air quality, increases the health risks such as respiratory illnesses, and will potentially contributes to greenhouse gas emissions.

ii. ***Impact on water*:** Water runoff from construction areas will contain debris and materials that can pollutes water bodies, which further threatens aquatic ecosystems and make the water unsuitable for human consumption.

3.1.4. Mechanical and Electronic Erection Activities

i. ***Impact on air*:** These activities will release noise and air pollutants from machinery, affecting the local air quality and stressing both humans and animals.

ii. ***Impact on water*:** Water runoff from erection areas, will be often containing oils and paints that introduces toxic pollutants into water bodies, contaminating the drinking water sources, harm the aquatic organisms, and cause a long-term ecological imbalance.

3.1.5. Influx of Labor and construction of Temporary Housing

i. ***Impact on land*:** The influx of labours and the construction of temporary housing for them can lead to a change in the land use pattern of the area and may lead to the destruction of the ecosystem and loss of habitat.

ii. ***Impact on water***: Sanitary effluents from labour camps can contaminate surface and groundwater sources, spreading diseases and posing health risks to nearby communities.

3.2. OPERATION PHASE

3.2.1. Impact on Land

i. **Landfills:** Deposition of unmanaged ash ponds and landfills results in land degradation, erosion, and contamination of surrounding soil through the leaching of heavy metals and chemicals.

ii. **Leachate Formation :**Leachate from fly ash ponds can contaminate soil and groundwater. This leachate might release hazardous heavy metals such as arsenic, lead, chromium, and cadmium

iii. **Dust Pollution**: There can be dust emissions when fly ash is dry disposed in landfills, which harm local air quality and contribute to soil and vegetation contamination when it settles. This dust can also be carried away with wind can cause damage to area which are away

3.2.2. Impact on air

i. **Coal combustion:** Due to coal combustion in thermal power plants significant quantities of air pollutants are released, including **particulate matter (PM)**, **sulfur dioxide (SO_2)**, **nitrogen oxides (NO_x)**, and **carbon dioxide (CO_2)**.

ii. **Particulate Matter (PM)**: Fly ash particles, which are emitted during coal combustion, contribute to airborne particulate matter, impacting human respiratory health and reducing air quality in surrounding areas.

iii. **Greenhouse Gas Emissions**: The burning of coal for power generation releases large amounts of **carbon dioxide (CO_2)**, contributing to climate change by increasing greenhouse gas concentrations in the atmosphere.

iv. **Sulfur Dioxide (SO_2) and Nitrogen Oxides (NO_x)** contributes to acid rain formation, which negatively affects both terrestrial and aquatic ecosystems by lowering soil and water pH.

v. **Health Impacts**: Airborne pollutants from coal plants are linked to respiratory diseases and other health problems in nearby populations

3.2.3. Impact on Water

i. **Surface Water (Rivers, Streams, and Water Bodies)**: Effluents from thermal power plants, including ash pond decants and wastewater from cooling systems, are often released into nearby rivers, streams, and water bodies. This increases turbidity and reduces primary productivity, harming aquatic life such as fish, plankton, and shellfish.

a. **Contamination with Heavy Metals**: Heavy metals like iron (Fe), manganese (Mn), lead (Pb), and mercury (Hg) from fly ash and effluents pollute surface waters. This can result in the contamination of the food chain and render the water unsafe for human consumption and aquatic organisms.

b. **Thermal Pollution**: The discharge from cooling towers into nearby water bodies raises water temperatures, which suppresses phytoplankton growth, reduces oxygen levels, and disrupts aquatic ecosystems.

c. **Eutrophication**: The elevated temperatures and increased nutrient load in water bodies can lead to eutrophication. As the Biological Oxygen Demand (BOD) increases and decreases, Dissolved Oxygen (DO) leads to a decline in water quality, further harming aquatic life.

d. **Water availability:** Due to large amount of withdrawal of water from the river the water availability for downstream users decreases. Natural flow of river can change as large amout of water is drawn from the river.

ii. **Groundwater:** The leachate from ash ponds and landfills can infiltrate the soil and contaminate groundwater with toxic metals like arsenic, cadmium, and chromium. This leads to groundwater contamination, making it unsafe for drinking and agricultural use.

a. **Leaching of Soluble Ions**: Acidic conditions (low pH) in ash ponds can increase the leaching of soluble ions and metals into groundwater, further exacerbating the contamination of local water sources (e.g., cadmium, chromium, and lead).

b. **Water Availability and Quality**: The overexploitation of groundwater for coal-based power plants can also lead to the depletion of groundwater levels, the drying up of wells, and saltwater intrusion into coastal areas.

3.2.4. Impact on Ecology

i. Withdrawal of large amounts of water can cause entrapment of aquatic organisms.

ii. This site is located about 4.5 to 5 km east of Western Ghats; changes in water quality, air pollution, and contamination of land from heavy metals can cause the breakdown of the fragile ecosystem of the Western Ghats

iii. The coal-based thermal power plants have a significant environmental footprint, which shows the requirement of robust mitigation measures and effective management practices to minimize their adverse impacts. The primary environmental impacts are having a cascading secondary effect. The poor air quality and noise pollution can lead to long-term health issues and will disturb the local wildlife. Soil contamination and loss of fertility of soil will reduce agricultural productivity. Contaminated water sources not only threaten aquatic life but also decrease the availability of safe drinking . The destruction of natural habitats and ecosystems will disrupt biodiversity and accelerate environmental degradation. These impacts highlight the interconnectedness of environmental parameters and underscores the need for mitigation measures to address both immediate and long-term consequences.

4. Possible Environmental Impact Mitigation Measure

To address the anticipated environmental impacts of the 1200 MW coal-based thermal power project, a range of mitigation measures must be implemented throughout the project phases.

4.1. OBJECTIVES

4.1.1. ***Reduce Adverse Environmental Effects***: Identify and reduce the negative impacts on air, water, soil, and biodiversity associated with the construction and operation of the power plant.

4.1.2. ***Protect Public Health and Safety***: Ensure the safety of local communities and plant workers by implementing controls on emissions, pollutants, and hazardous materials.

4.1.3. ***Promote Sustainable Resource Use***: Optimize resource use (water, energy, land) to limit waste and environmental degradation.

4.1.4. ***Ensure Compliance with Environmental Standards***: Adhere to all national and regional environmental regulations, international standards, and best practices.

4.1.5. ***Foster Community Engagement***: Address community concerns through regular communication, transparency, and grievance mechanisms.

4.2. METHODOLOGY FOR DEVELOPING AND IMPLEMENTING MITIGATION ACTIONS

4.2.1. Baseline Environmental Assessment

i. ***Data Collection***: Conduct a thorough baseline study of current environmental conditions, including air and water quality, soil characteristics, noise levels, local flora and fauna, and socio-economic conditions.

ii. ***Identifying Sensitive Areas***: Map out ecologically sensitive areas, local water bodies, and nearby communities to identify potential high-impact zones.

4.2.2. Environmental Impact Identification and Analysis

i. ***Impact Prediction***: Use predictive modeling to assess potential environmental impacts across various project phases (pre-construction, construction, and operation).

ii. ***Impact Analysis***: Evaluate the significance of these impacts in terms of their magnitude, duration, reversibility, and geographical scope, which provides the foundation for tailored mitigation actions.

4.2.3. Stakeholder Engagement and Community Consultation

i. ***Consult with Local Stakeholders***: Engage local communities, authorities, and environmental groups to discuss anticipated impacts, receive input, and gather local knowledge to enhance mitigation measures.

ii. ***Transparency and Grievance Mechanisms***: Establish communication channels for stakeholders to voice concerns or provide feedback on mitigation strategies.

4.2.4. Development of Mitigation Strategies

i. ***Best Available Technologies (BATs)***: Identify and incorporate BATs and industry best practices to minimize emissions, water use, waste generation, and noise pollution.

ii. ***Site-Specific Mitigation Plans***: Tailor actions to the specific environmental and social context of the Vile Bhagad MIDC area to effectively address site-specific impacts.

4.2.5. Monitoring, Review, and Adaptive Management

i. ***Establish Monitoring Protocols***: Define key performance indicators (KPIs) and monitoring schedules to assess the effectiveness of mitigation measures over time.

ii. ***Adaptive Management***: Use monitoring data to refine and adjust mitigation strategies as needed to ensure long-term compliance and environmental protection.

4.3. CONSTRUCTION AND PRECONSTRUCTION PHASE

4.3.1. Air Quality Management

Greenbelt Development: Create ***a buffer zone of native vegetation*** around the site. Trees and shrubs act as natural barriers, absorbing CO_2 and reducing dust dispersion, which can improve local air quality. The trees selected should be so chosen that they provide multifarious uses, which can also be used for the socio-economic upliftment of the local communities. E.g.: Peepal (*Ficus religiosa*), Khair (*Acacia catechu*)

4.3.2. Water Resource Management

Effluent Treatment Plant (ETP): Install ETPs for treating wastewater before discharge. Treated water can be recycled back into the cooling system, reducing the demand for fresh water.

4.3.3. Noise Pollution Control

i. ***Community Communication***: Regularly inform nearby residents about noise levels and incorporate feedback on noise reduction. Feedback must be incorporated into the project.

ii. ***Noise Barriers and Vegetation Buffers***: Construct noise barriers and plant dense vegetation (Shelterbelts) around the perimeter to absorb noise. This will also increase forest cover and provide environmental benefits.

4.3.4. Soil and Waste Management

i. ***Topsoil Conservation***: During construction, preserve the topsoil for future site reclamation or landscaping activities, helping maintain soil quality for agriculture.

ii. ***Spill Prevention and Soil Protection***: Construct containment systems for coal and ash storage to prevent spills or seepage. Regularly monitor soil around the plant to detect contaminants and ensure swift remediation if necessary.

4.3.5. Biodiversity Conservation

i. ***Seasonal Restrictions***: Restrict construction during critical breeding seasons or migratory periods for local species for sensitive habitats are identified nearby.

ii. ***Monitoring and Conservation Offsetting***: Set up biodiversity monitoring programs and, where impacts are unavoidable, consider conservation offsets to protect habitats or species in nearby areas.

4.4.OPERATION PHASE

4.4.1. Air Quality Management

i. ***Emission Reduction Technologies***: Install ***flue gas desulfurization (FGD)*** systems to reduce ***sulfur dioxide (SO_2) emissions***, ***selective catalytic reduction (SCR)*** units for ***nitrogen oxides (NOx)***, and ***electrostatic precipitators (ESP)*** for particulate matter.

ii. ***Dust Control in Coal Handling***: Enclose coal handling and storage facilities to reduce fugitive dust. Use ***water spray systems*** and ***dust suppressants*** in storage and transportation areas.

4.4.2. Water Resource Management

i. ***Closed-Loop Cooling System***: Implement ***closed-loop cooling*** to reduce water withdrawal from local sources (river), ensuring water availability for other communities and agricultural needs.

ii. ***Thermal Pollution Mitigation***: Cool discharge water to safe levels before release to prevent thermal pollution that could affect local aquatic ecosystems in the river.

iii. ***Leachate Prevention in Ash Ponds***: Line ash ponds with impermeable materials and construct leachate collection systems to prevent groundwater contamination.

4.4.3. Soil and Waste Management

i. ***Fly Ash Utilization***: Implement fly ash utilization strategies, such as using it in construction materials and road construction, to minimize disposal volumes.

ii. ***Post-Project Site Restoration***: Develop a site reclamation plan to restore the land, reintroduce vegetation, and rehabilitate soil quality once the project concludes.

iii. ***Spill Prevention and Soil Protection***: Construct containment systems for coal and ash storage to prevent spills or seepage. Regularly monitor soil around the plant to detect contaminants and ensure swift remediation if necessary.

4.4.4. Noise Pollution Control

i. ***Soundproofing and Maintenance***: Priority should be placed on minimizing noise generation, and if that is not possible to achieve, then soundproof equipment like turbines should be used, and regular maintenance should be conducted to reduce operational noise.

ii. ***Community Communication***: Regularly inform nearby residents about noise levels and incorporate feedback on noise reduction. Feedback must be incorporated into the project.

4.4.5. Biodiversity Conservation

Buffer Zones and Wildlife Corridors: Establish buffer zones around ecologically sensitive areas and create wildlife corridors to reduce habitat fragmentation. This will allow peaceful coexistence, although disturbed.

4.5.POTENTIAL MANAGEMENT PLANS

4.5.1. Environmental Management Plan (EMP)

i. Outline protocols for waste management, air and water pollution controls, noise mitigation, and community engagement.

ii. Define responsibilities, timelines, and reporting requirements for each mitigation action.

4.5.2. Safety, and Environmental (HSE) Management Plan

- ***i.*** Ensure worker and community safety through protocols for managing pollutants, hazardous materials, and emergency responses.

- ***ii.*** Conduct safety training for employees and maintain clear communication channels for the public.

4.5.3. Biodiversity Management Plan

- ***i.*** Develop specific measures to protect local flora and fauna, ensuring compliance with biodiversity regulations.

- ***ii.*** Include monitoring protocols to detect changes in species populations and habitats over time.

4.5.4. Waste Management Plan

- ***i.*** Detail how solid and liquid wastes, including fly ash and effluents, will be safely treated, recycled, or disposed of.

- ***ii.*** Include protocols for handling hazardous waste in compliance with the Hazardous Waste Management Rules.

4.5.5. Community Engagement and Social Impact Management Plan

- ***i.*** Establish clear communication channels and grievance mechanisms for local communities to report concerns.

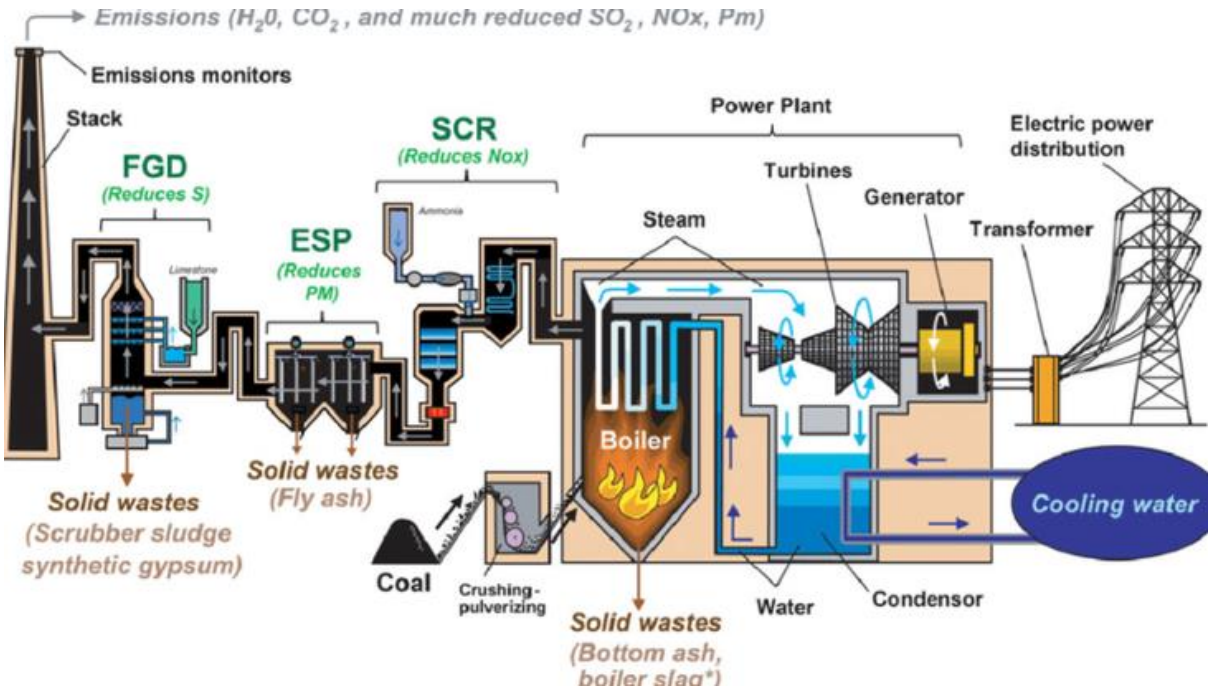
- ***ii.*** Regularly update communities on mitigation progress and involve them in environmental stewardship efforts, like greenbelt development.

4.5.6. Emergency Preparedness and Response Plan

- ***i.*** Develop a response plan to address potential emergencies, such as chemical spills, fires, or equipment malfunctions.

- ***ii.*** Conduct drills, establish emergency teams, and communicate response procedures with local authorities and communities.

"Implementing these mitigation measures and a robust environmental monitoring and management plan will help minimize the environmental and social impacts of the proposed 1200 MW coal-based thermal power project at Vile Bhagad MIDC."



Pic 2. Visualized model of mitigation measures in thermal power plant

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